departments Table:

department_id	department_name
1	Human Resources
2	Finance
3	Sales
4	Marketing
5	Engineering
6	IT Support
7	Operations
8	Product Management
9	Customer Service
10	Legal

employees Table:

employee_id	first_name	last_name	salary	hire_date	department_id
1	Amit	Sharma	60000	2015-03-25	1
2	Priya	Patel	70000	2016-07-10	2
3	Raj	Kumar	55000	2018-02-15	3
4	Anita	Reddy	75000	2014-11-01	4
5	Vikram	Singh	90000	2017-09-20	5
6	Neha	Verma	95000	2019-01-22	6
7	Ravi	Gupta	80000	2015-06-18	7
8	Sita	Iyer	68000	2018-05-30	8
9	Kiran	Mehta	71000	2020-03-05	9
10	Manoj	Nair	50000	2016-12-10	10

Medium:

1. Retrieve the top 3 highest-paid employees from each department.
2. Write a query to find the department with the highest number of employees.
3. List all the employees whose salary is higher than the average salary of their respective departments.
4. Write a query that calculates the cumulative sum of salaries by department, ordered by the employee's hire date.
5. Write a query to find employees who work in departments where the average salary is greater than the overall company average salary.
6. How would you find the second-highest salary in the company without using `LIMIT` or `TOP`?
7. Write a query to find the most recent department each employee has worked in.
8. Write a query to find departments that have no employees.
9. Write a query to find employees who joined the company within the last 30 days.

Hard:

1. Write a query to retrieve the employees who have received the highest salary in their department.

2. Write a query to find employees whose salary is higher than the average salary of all employees, but only if their department's average salary is also above the company average salary.
3. Given a table of employee salaries, write a query to find the difference between the highest and the lowest salary in each department.
4. Write a query to find employees who earn more than the average salary in their department and the department's average salary is greater than the company's average salary.
5. Write a query to find the departments with employees who have worked more than 10 years, and those departments should have at least 5 employees.
6. Write a query to return the number of employees in each department, including departments with no employees.
7. Write a query to find the employee(s) who have worked for the longest period in the company, along with their department name.
8. Find the total salary expenditure for each department and list the department(s) with the maximum total salary.
9. Write a query to find all employees who earn the same salary as at least one other employee in the company.
10. Write a query to find employees who have been with the company the longest in terms of service and salary above the average salary of the company.

2. Advanced SQL Joins:

Medium:

1. Write a query to find all employees who have worked in multiple departments.
2. Retrieve a list of employees who are not assigned to any department.
3. Write a query to find the departments and employees where the department name starts with 'S'.
4. Write a query to find the employee(s) with the highest salary in each department using `JOIN`.
5. Retrieve all employees who have worked in the same department for more than 3 years.
6. Write a query to find the number of employees working under each manager, including managers with no employees.
7. Find the departments that have the most employees, including those with ties.
8. Retrieve the employees who do not have a manager assigned to them.
9. Write a query to find employees who are in the same department as the employee with the highest salary.
10. Write a query that joins three tables: Employees, Departments, and Salaries, to find employees with salaries above the average salary in their department.

Hard:

1. Write a query to find all employees who have been in the same department for more than 2 years and their salary is higher than the department average.
2. Retrieve a list of employees and their managers who have been with the company for more than 5 years.
3. Write a query to find all employees who have worked in more than one department and their total salary across departments.

4. Write a query to retrieve employees who belong to the same department and have similar salaries (within a 5% range).
5. Write a query to return the department that has the highest total salary expenditure, including the department name and the total salary.
6. Write a query that finds all employees who belong to the same department as an employee whose salary is in the top 10 highest in the company.
7. Retrieve all departments where the sum of employee salaries is greater than the average salary of all employees.
8. Write a query to find employees whose salaries exceed the average salary in their department and who are managed by a specific manager.
9. Find the top three employees in each department who have the highest salary.
10. Retrieve a list of all employees who have the same job title but work in different departments.

3. SQL Subqueries:

Medium:

1. Write a query to find employees who earn more than the average salary of all employees in the company using a subquery.
2. Write a query to find departments where the average salary is greater than the salary of the employee with the lowest salary.
3. Write a query to find employees whose salary is higher than the average salary in the department, using a subquery in the `WHERE` clause.
4. Write a query to find the employee(s) with the highest salary in each department using a subquery.
5. Write a query to find the employee(s) who earn more than the employee with the second-highest salary in the company.
6. Write a query to find the department(s) with the most employees, including those departments with ties.
7. Write a query to find the average salary of employees in departments with at least 10 employees.
8. Write a query to find all employees who have been with the company for longer than the average tenure of all employees.
9. Write a query that returns employees whose salary is above the department's average salary, using a correlated subquery.
10. Find employees who have the same salary as the employee with the highest salary in the company using a subquery.

Hard:

1. Write a query to find employees who earn more than the average salary in their department and whose department's average salary is higher than the overall company’s average salary.
2. Write a query to find the second-highest salary in each department using a subquery.
3. Write a query that retrieves the department(s) with the highest total salary expenditure using a subquery.

4. Write a query to find all employees who earn more than 50% of the employees in the company using a subquery.
5. Write a query that returns the names of departments whose average salary is higher than the salary of the employee with the lowest salary.
6. Write a query to find all employees who earn more than the highest-earning employee in their department.
7. Write a query to find employees who belong to a department where the total salary is above the company average.
8. Write a query to find departments with more than 10 employees, where the average salary of employees is greater than the average salary of the company.
9. Write a query to find the employee(s) whose salary is greater than the average salary in their department and the salary of the lowest-paid employee in the company.
10. Write a query that returns the name of the department where the average salary of employees exceeds the company-wide average salary.